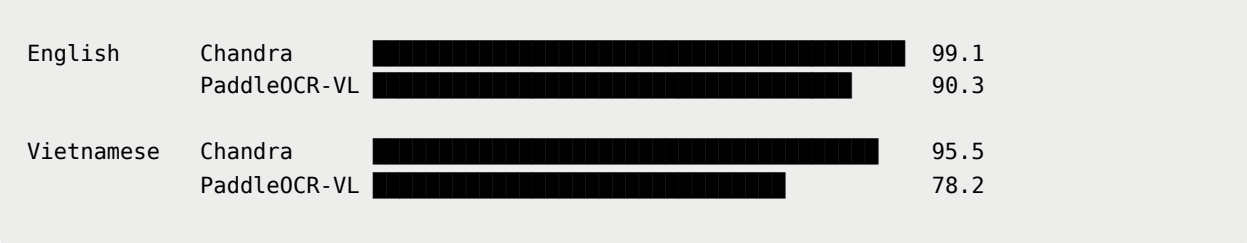
notion_handwriting_results

Handwriting OCR — what works, what breaks

Chandra OCR 2 · synthetic medical forms · metric = share of handwritten fields correctly found · paired reference in every run

1. Baseline, clean pages

model	English	Vietnamese
Chandra OCR 2 (5 B)	**99.1%**	**95.5%**
PaddleOCR-VL (0.9 B)	90.3%	78.2%



2. Degradation — the headline

24 levels × 3 specimens, English. Reference `niveau0` = **100.0%**.

degradation	range tested	worst score	verdict
sensor noise	500 → 8 photons	**100.0%**	no effect
underexposure	−2 → −6 stops	**100.0%**	no effect
JPEG	quality 25 → 3	**97.2%**	negligible
motion blur	6 → 37 px	**8.5%**	breaks
resolution	1241 → 364 px	**18.3%**	breaks

```
worst score reached over the whole tested range

noise       ████████████████████████████████  100.0
exposure    ████████████████████████████████  100.0
jpeg        ███████████████████████████████    97.2
resolution  ██████                             18.3
blur        ██                                  8.5
```

Why photometry costs nothing — noise, darkness and compression change pixel *values*. The stroke stays where it is, and its shape is what carries the letter.

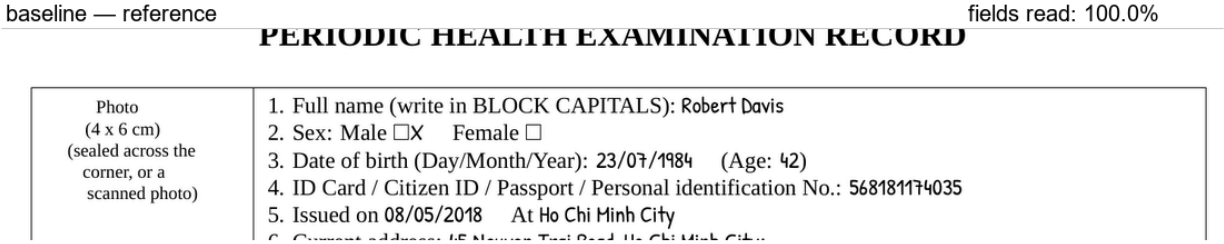
Why geometry breaks it — blur and downscaling change stroke *shape*. Cursive strokes are thin and close together; once two of them merge, no amount of context recovers the character.

---

3. Motion blur

blur	% of page width	at 200 dpi	fields found
6 px	0.36%	0.8 mm	**97.2%**
11 px	0.67%	1.4 mm	73.2%
17 px	1.03%	2.2 mm	43.7%
24 px	1.45%	3.0 mm	15.5%
37 px	2.24%	4.7 mm	8.5%

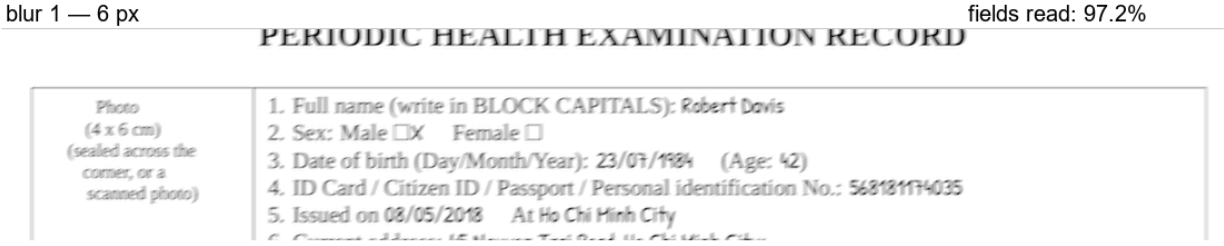
baseline — reference fields read: 100.0%



PERIODIC HEALTH EXAMINATION RECORD

Photo (4 x 6 cm) (sealed across the corner, or a scanned photo)	1. Full name (write in BLOCK CAPITALS): Robert Davis 2. Sex: Male ☐X Female ☐ 3. Date of birth (Day/Month/Year): 23/07/1984 (Age: 42) 4. ID Card / Citizen ID / Passport / Personal identification No.: 568181174035 5. Issued on 08/05/2018 At Ho Chi Minh City 6. Current address: 45 Nguyen Trai Road, Ho Chi Minh City

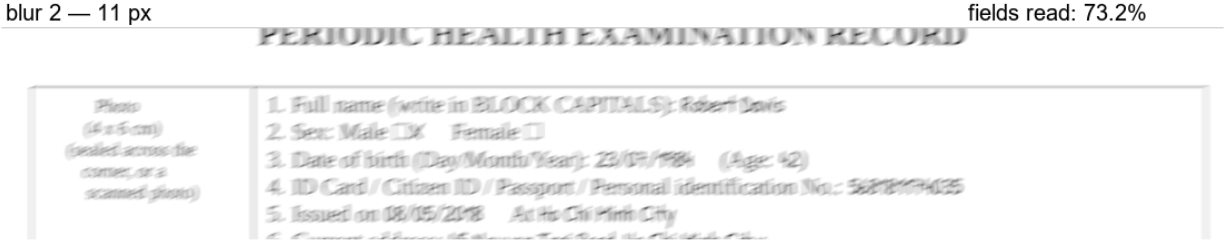
blur 1 — 6 px fields read: 97.2%



PERIODIC HEALTH EXAMINATION RECORD

Photo (4 x 6 cm) (sealed across the corner, or a scanned photo)	1. Full name (write in BLOCK CAPITALS): Robert Davis 2. Sex: Male ☐X Female ☐ 3. Date of birth (Day/Month/Year): 23/07/1984 (Age: 42) 4. ID Card / Citizen ID / Passport / Personal identification No.: 568181174035 5. Issued on 08/05/2018 At Ho Chi Minh City 6. Current address: 45 Nguyen Trai Road, Ho Chi Minh City

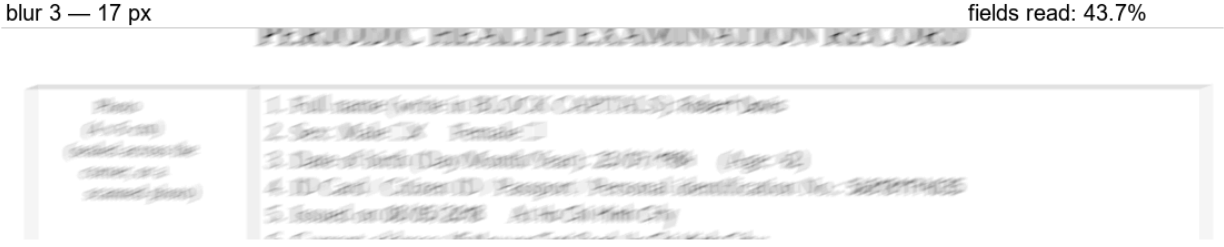
blur 2 — 11 px fields read: 73.2%



PERIODIC HEALTH EXAMINATION RECORD

Photo (4 x 6 cm) (sealed across the corner, or a scanned photo)	1. Full name (write in BLOCK CAPITALS): Robert Davis 2. Sex: Male ☐X Female ☐ 3. Date of birth (Day/Month/Year): 23/07/1984 (Age: 42) 4. ID Card / Citizen ID / Passport / Personal identification No.: 568181174035 5. Issued on 08/05/2018 At Ho Chi Minh City 6. Current address: 45 Nguyen Trai Road, Ho Chi Minh City

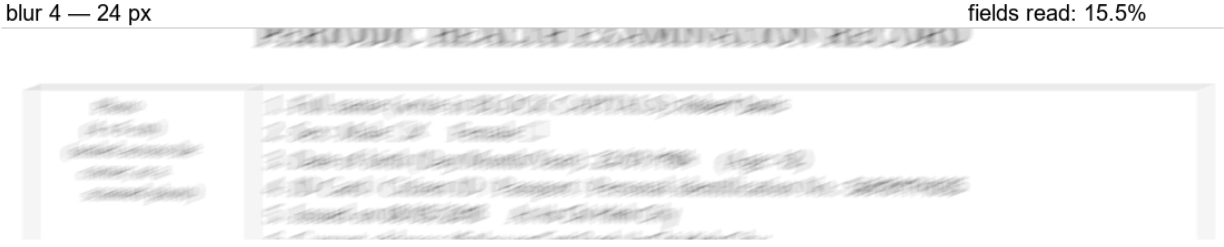
blur 3 — 17 px fields read: 43.7%



PERIODIC HEALTH EXAMINATION RECORD

Photo (4 x 6 cm) (sealed across the corner, or a scanned photo)	1. Full name (write in BLOCK CAPITALS): Robert Davis 2. Sex: Male ☐X Female ☐ 3. Date of birth (Day/Month/Year): 23/07/1984 (Age: 42) 4. ID Card / Citizen ID / Passport / Personal identification No.: 568181174035 5. Issued on 08/05/2018 At Ho Chi Minh City 6. Current address: 45 Nguyen Trai Road, Ho Chi Minh City

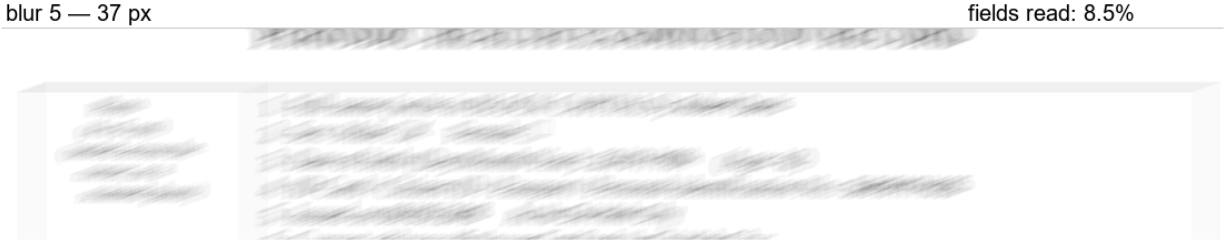
blur 4 — 24 px fields read: 15.5%

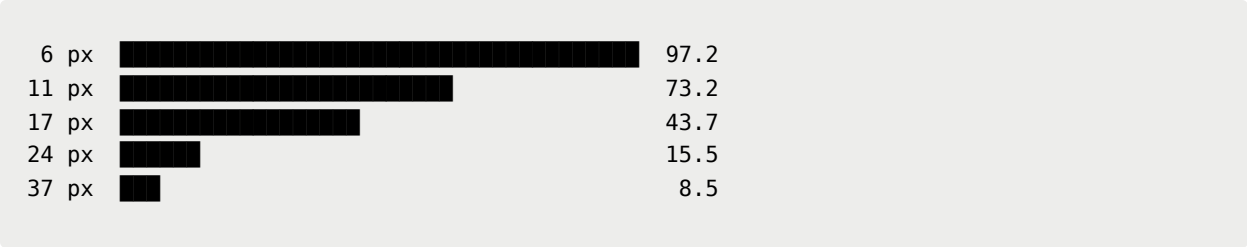


PERIODIC HEALTH EXAMINATION RECORD

Photo (4 x 6 cm) (sealed across the corner, or a scanned photo)	1. Full name (write in BLOCK CAPITALS): Robert Davis 2. Sex: Male ☐X Female ☐ 3. Date of birth (Day/Month/Year): 23/07/1984 (Age: 42) 4. ID Card / Citizen ID / Passport / Personal identification No.: 568181174035 5. Issued on 08/05/2018 At Ho Chi Minh City 6. Current address: 45 Nguyen Trai Road, Ho Chi Minh City

blur 5 — 37 px fields read: 8.5%



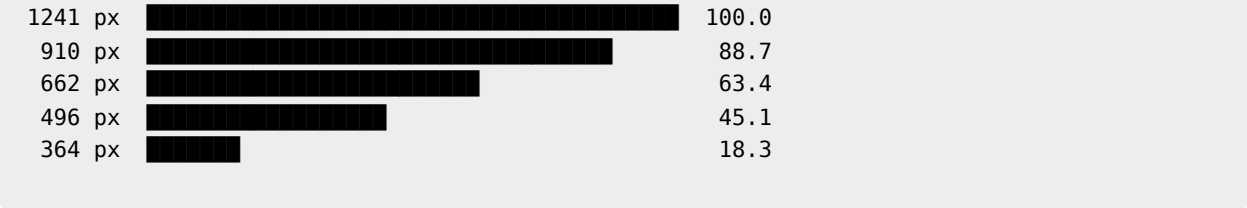


≈ **25 points lost per step** — a steep slope, not a cliff. A blurred page degrades gradually, so a confidence signal has something to work with.

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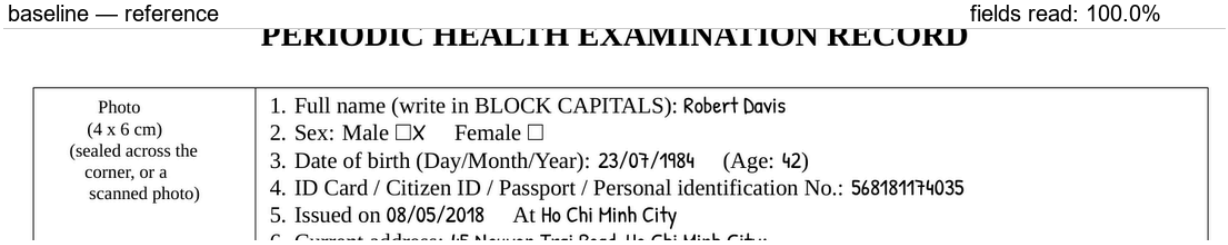
4. Resolution

page width	approx. dpi	fields found
1241 px	150	**100.0%**
910 px	110	88.7%
662 px	80	63.4%
496 px	60	45.1%
364 px	44	18.3%



baseline — reference

fields read: 100.0%

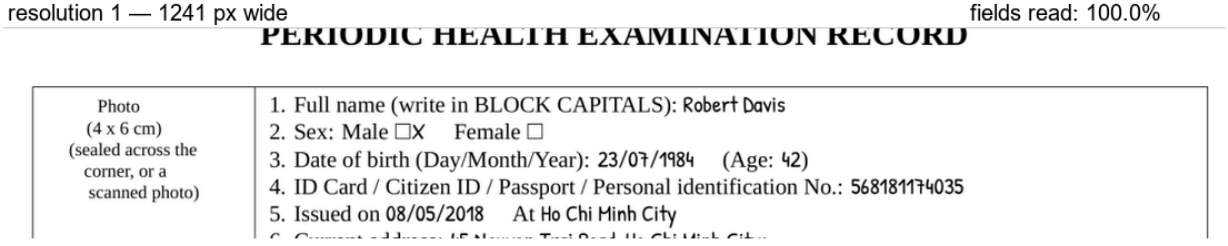


PERIODIC HEALTH EXAMINATION RECORD

Photo (4 x 6 cm) (sealed across the corner, or a scanned photo)	1. Full name (write in BLOCK CAPITALS): Robert Davis 2. Sex: Male ☐X Female ☐ 3. Date of birth (Day/Month/Year): 23/07/1984 (Age: 42) 4. ID Card / Citizen ID / Passport / Personal identification No.: 56818117 4035 5. Issued on 08/05/2018 At Ho Chi Minh City

resolution 1 — 1241 px wide

fields read: 100.0%

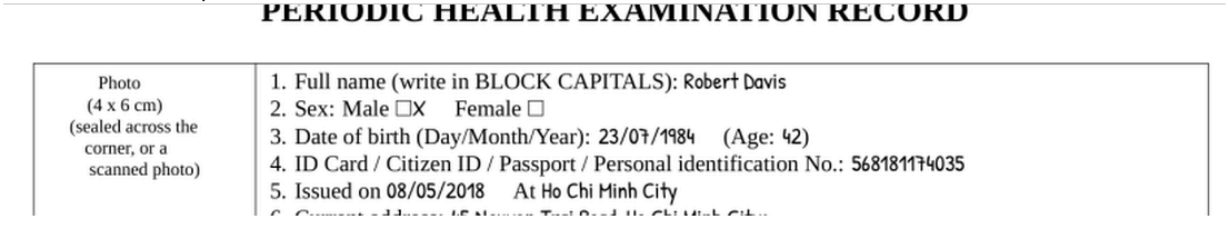


PERIODIC HEALTH EXAMINATION RECORD

Photo (4 x 6 cm) (sealed across the corner, or a scanned photo)	1. Full name (write in BLOCK CAPITALS): Robert Davis 2. Sex: Male ☐X Female ☐ 3. Date of birth (Day/Month/Year): 23/07/1984 (Age: 42) 4. ID Card / Citizen ID / Passport / Personal identification No.: 56818117 4035 5. Issued on 08/05/2018 At Ho Chi Minh City

resolution 2 — 910 px

fields read: 88.7%

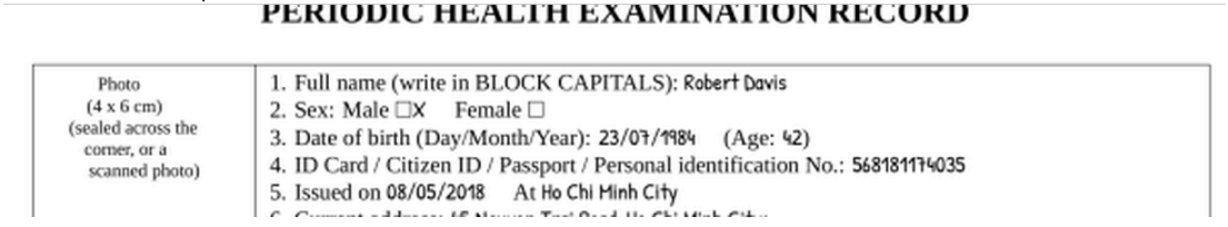


PERIODIC HEALTH EXAMINATION RECORD

Photo (4 x 6 cm) (sealed across the corner, or a scanned photo)	1. Full name (write in BLOCK CAPITALS): Robert Davis 2. Sex: Male ☐X Female ☐ 3. Date of birth (Day/Month/Year): 23/07/1984 (Age: 42) 4. ID Card / Citizen ID / Passport / Personal identification No.: 56818117 4035 5. Issued on 08/05/2018 At Ho Chi Minh City

resolution 3 — 662 px

fields read: 63.4%

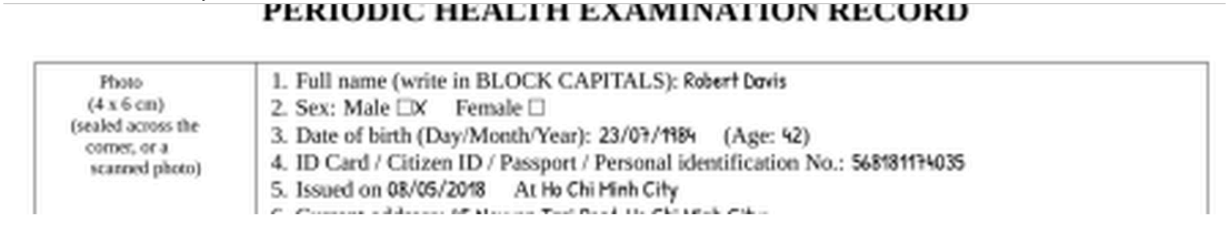


PERIODIC HEALTH EXAMINATION RECORD

Photo (4 x 6 cm) (sealed across the corner, or a scanned photo)	1. Full name (write in BLOCK CAPITALS): Robert Davis 2. Sex: Male ☐X Female ☐ 3. Date of birth (Day/Month/Year): 23/07/1984 (Age: 42) 4. ID Card / Citizen ID / Passport / Personal identification No.: 56818117 4035 5. Issued on 08/05/2018 At Ho Chi Minh City

resolution 4 — 496 px

fields read: 45.1%



PERIODIC HEALTH EXAMINATION RECORD

Photo (4 x 6 cm) (sealed across the corner, or a scanned photo)	1. Full name (write in BLOCK CAPITALS): Robert Davis 2. Sex: Male ☐X Female ☐ 3. Date of birth (Day/Month/Year): 23/07/1984 (Age: 42) 4. ID Card / Citizen ID / Passport / Personal identification No.: 56818117 4035 5. Issued on 08/05/2018 At Ho Chi Minh City

resolution 5 — 364 px

fields read: 18.3%



PERIODIC HEALTH EXAMINATION RECORD

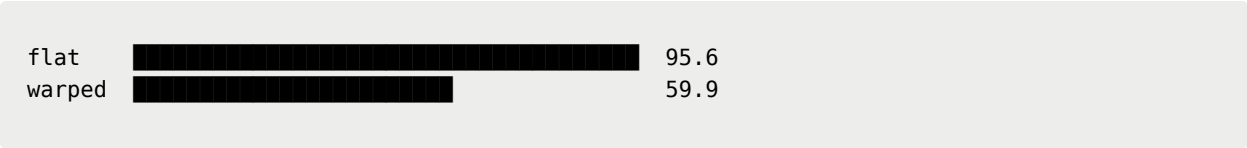
Photo (4 x 6 cm) (sealed across the corner, or a scanned photo)	1. Full name (write in BLOCK CAPITALS): Robert Davis 2. Sex: Male ☐X Female ☐ 3. Date of birth (Day/Month/Year): 23/07/1984 (Age: 42) 4. ID Card / Citizen ID / Passport / Personal identification No.: 56818117 4035 5. Issued on 08/05/2018 At Ho Chi Minh City

A 25% downscale from the source costs **nothing**. The model does not need 200 dpi — it needs about half of it.

5. 3D deformation

Vietnamese specimens projected on curved meshes, vs the same set read flat.

condition	fields found
flat (30 specimens)	**95.6%**
warped (8 specimens)	**59.9%**



−35.7 points — the largest single drop measured.

> ⚠️ Lower bound only: those 8 renders came out rotated 90° (texture orientation bug found afterwards), so the figure mixes deformation with orientation.

6. Capture specification

To keep **≥ 90% of fields**, on English handwriting, with Chandra:

requirement	threshold
motion blur	< 7 px at 200 dpi — **0.4% of page width, ≈ 0.9 mm on A4**
resolution	≥ **950 px across the page** (≈ 115 dpi)
noise · exposure · JPEG	**no constraint** within tested ranges

→ In practice: **hold still, don’t let the image be downscaled.**
Lighting is not the problem.